BT5153 Applied Machine Learning for Business Analytics

Customer Review Analysis for Product Insights

Group 4 Project Proposal

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Intent

In the contemporary electronics market, star ratings alone are insufficient; they indicate satisfaction but mask the underlying drivers of sentiment. Faced with millions of unstructured reviews, businesses often overlook specific pain points that cause returns or drive loyalty. Our project transforms such "dark data" into actionable insights, revealing where products excel or fail. By converting feedback into structured data, this scalable, cost-effective approach monitors performance without manual audits. Topic modeling identifies recurring complaints, enabling targeted improvements in hardware or software. Ultimately, this analysis proves that sentiment-driven decisions align product strategies with real user needs, moving beyond surface-level metrics.

Data Source and Scope

The analysis uses the Amazon V2 dataset, specifically the Electronics category, to ensure relevance and sufficient volume. From the initial pool of 22 million reviews across 787k products, we will extract five key fields: rating (1–5), verification status, product ID (asin), review text, and summary. To maintain maximum data coverage, the complete dataset will be used instead of pre-filtered subsets.

To refine the dataset, we will apply the following filters:

- **Noise Reduction:** Remove reviews with fewer than 20 words.
- **Statistical Significance:** Exclude products with fewer than 200 reviews.
- **Balance:** Cap at 500 reviews per product to prevent bias from popular items.

Post-cleaning, the final dataset is expected to consist of approximately 4 million reviews covering 11k products.

Methodology and Deliverables

Text Processing and Feature Extraction: Customer reviews are cleaned to remove noise and irrelevant content, then tokenized and transformed into features using TF-IDF and pre-trained embeddings (e.g., BERT) to capture semantic meaning. Key fields such as product IDs and timestamps are retained for analysis.

Topic Extraction: Reviews are analyzed using topic modeling techniques such as LDA or BERTopic to identify recurring themes. This enables the extraction of key issues and discussion patterns present in customer feedback.

Insight Generation and Business Interpretation: Extracted topics are aggregated to generate product-level insights, including frequent complaint areas and common user concerns. Results are presented through visualizations or a dashboard to support data-driven product improvements.

Milestones

Phase	Timeline/Weeks	Key Deliverables
Data Collection	1	Amazon Electronics dataset (selected fields: rating, verified, asin, reviewText, summary)

Preprocessing & EDA	1	Cleaned dataset, filtered products, capped reviews, exploratory analysis (rating distribution, verified patterns)
Model Development	2–3	Sentiment classification, aspect extraction (topics/keywords), feature engineering
Evaluation & Tuning	4	Model comparison (LDA and BERT), performance metrics (coherence, topic diversity and Interpretation), fine-tuning
Report & Presentation	5	Final report, dashboard prototype (Tableau/Power BI), presentation slides

Constraints

- **Offline Analysis Only:** The system processes historical customer reviews, not real-time streaming or monitoring data.
- **Reliance on Pre-trained Models:** Due to computational constraints, we use pre-trained models (e.g., VADER, pre-trained embeddings) instead of training full deep learning models from scratch.
- **Associative Findings Only:** The analysis identifies patterns and associations, but does not establish causation between product features and ratings.
- **English Language Restriction:** The project is limited to English-language reviews to ensure consistent NLP processing.
- **Data Balancing:** We enforce balanced product representation by filtering products (min 200 reviews) and capping reviews (max 500 per product). This ensures meaningful analysis but may exclude certain edge-case insights.
- **Aspect extraction limitations:** Product features are inferred using unsupervised methods (e.g., topic modeling), which may not always result in perfectly standardized or interpretable aspects.

Desired Outcome: Success Metrics

1. **Topic Interpretability:** Identify distinct, meaningful product themes from negative reviews, measured by coherence and business relevance.
2. **RAG Accuracy:** Produce grounded, factually consistent review summaries that minimize errors and capture primary sentiment drivers.
3. **Insight Aggregation:** Consolidate sentiment distributions and recurring complaints into high-level comparisons across the product catalog.
4. **Business Intelligence Value:** Ensure the pipeline provides deeper diagnostic insights and actionable improvements than standard star-rating metrics.